Biomechanical and Clinical Deconstruction of Open-Footwear Sensor Systems in Diabetic Plantar Care

The absolute assertion that every published smart insole requires an enclosed shoe, and that no scientific literature, patent, or commercial product has ever addressed sandal-wearing populations such as India's diabetic demographic, is demonstrably false. While public health literature confirms that open sandals, slippers, and chappals constitute the primary footwear across the Indian subcontinent[1], biomedical engineers, clinical podiatrists, and medical device companies have explicitly developed, patented, and clinically validated sensor-instrumented insoles and smart systems designed specifically for open footwear[2].

The technical literature reveals that the predominance of closed-shoe systems in Western commercial markets does not reflect an absolute engineering impossibility, but rather an artifact of regional market prioritization[1]. To address the unique biomechanical and climatic challenges of tropical populations, academic institutions globally and within India have formulated dedicated sensorized sandals, flexible optoelectronic arrays, piezoresistive nanocomposites, and dynamically responsive metamaterials[2].

Empirical Refutation: Published Instrumented Sandal and Open-Footwear Sensor Systems

A systematic review of academic databases directly refutes the claim that smart insoles are exclusively confined to closed shoes. Multiple peer-reviewed studies document flexible sensing footbeds, force-sensitive resistor (FSR) arrays, and optical matrices fabricated within open sandals to monitor ambulatory plantar mechanics and mitigate diabetic ulcer risk[2].

In work published in *Sensors*, Safarloo and colleagues designed and experimentally validated an ambulatory plantar pressure monitoring platform relying on macro-bending attenuation in polymer optical fibers (POFs)[3]. The authors explicitly circumvented the geometry of closed shoes by mounting their flexible optical sensor network directly into the insole of an open sandal[3]. By tracking light intensity modulation induced by curvature deformation during ground contact, the system captures localized kinetic force beneath the calcaneus and metatarsal heads without encasing the dorsal aspect of the foot[3]. The authors contextualized the instrumented sandal specifically as a non-invasive biofeedback and prophylactic diagnostic tool for diabetic foot ulceration[3].

In an investigation published in *Wearable Technologies*, Burch and co-workers developed an insole sensing platform based on carbon nanotube (CNT) coated textile nanocomposites[4]. The investigators embedded these piezoresistive textile sensors across the forefoot and hindfoot regions of an open sandal footbed to quantify resultant ground reaction forces and dynamic foot-ground interactions[4]. By validating the system against laboratory-grade instrumented treadmills across multiple walking velocities, the authors established that flexible textile sensors

mounted on open sandals achieve measurement fidelity comparable to enclosed smart footwear, while offering superior breathability and substantial cost reductions[4].
Similarly, Ozturksoy and colleagues documented an instrumented sandal engineered with a 3D-printed thermoplastic polyurethane (TPU) insole combining six FSR pads and three non-anatomically routed TPU optical fiber sensors[2]. Specifically targeting pathological gait analysis and diabetic foot biomechanics, the researchers demonstrated that supervised machine learning regressors—including Adaptive Boosting, Gradient Boosting, and artificial neural networks—could decode optical attenuation signals to estimate continuous plantar forces with high correlation coefficients (R^2 between 0.865 and 0.951) and mean absolute errors within 5% to 7% of full scale[2]. The mechanical chassis supporting this entire sensing and telemetry architecture was a multi-strap open sandal[2].
Alternative open-source biomechanical sensing platforms have deliberately abandoned conventional enclosed insoles to serve low-resource clinics. Proulx et al. developed an articulated pedobarographic system featuring 704 individual piezoresistive sensing cells distributed across an interlocking matrix of rigid-linkage treads[8]. Designed using VeroClear photopolymers to emulate injection-molded polymers, the articulated bed provides an unyielding mechanical backing capable of supporting loads up to 1,000 pounds while flexing naturally along the longitudinal axis of an open sandal sole[8]. The authors emphasized that reliance on traditional enclosed insoles severely restricts clinical diagnostics in developing countries, configuring their platform to operate directly within accessible sandal bases[8].
Further supporting this paradigm, rehabilitative gait literature reviewed in *Frontiers in Neurology* highlights wearable systems such as that engineered by Santoso and Setyanto, which incorporates dual force-sensing resistors directly into the front and rear contact interfaces of a sandal insole to capture kinetic gait temporal parameters wirelessly[10].

Research Platform and Authorship	Sensor Transduction Modality	Footwear Chassis	Primary Clinical or Biomechanical Target
Safarloo et al., MDPI *Sensors*[3]	Polymer Optical Fiber (POF) macro-bend attenuation	Open sandal footbed with PP/EVA substrata[3]	Ambulatory plantar pressure monitoring and diabetic ulcer prophylaxis[3]
Burch et al., *Wearable Technologies*[4]	Carbon nanotube (CNT) coated textile piezoresistors	Open sandal footbed across forefoot and hindfoot[4]	Ground reaction force estimation and dynamic loading analysis[4]

Ozturksoy et al., *Int. J. 3D Print. Technol.* [2]	3D-printed TPU optical fibers + 6 FSR reference pads	Multi-strap instrumented open sandal[2]	Real-time plantar force estimation via supervised machine learning[2]
Proulx et al., Hilaris Open Access [8]	704-cell piezoresistive matrix on interlocking treads	Articulated linkage embedded in a sandal base[8]	Open-source, low-resource pedobarography for low-income settings[8]
Santoso & Setyanto, *Frontiers in Neurology* [10]	Dual piezoresistive FSRs with RF telemetry	Sandal insole with anterior and posterior contacts[10]	Wearable gait phase identification and kinetic footfall tracking[10]

The Indian Clinical Reality: Epidemiological and Biomechanical Determinants

The premise of the query regarding user demographics is clinically sound: India possesses one of the world's largest diabetic populations, exceeding 77 to 100 million diagnosed individuals, of whom approximately 50% are vulnerable to diabetic peripheral neuropathy (DPN) and subsequent foot complications[1]. However, the assumption that research communities have overlooked this population's footwear habits ignores decades of podiatric public health scholarship[1].

International clinical guidelines, such as those issued by the International Working Group on the Diabetic Foot (IWGDF), historically advocated for extra-depth, rigid-soled enclosed shoes equipped with custom orthoses to immobilize digital deformities, shield insensate skin from exogenous puncture wounds, and offload excessive vertical forces[17]. While effective in temperate Western climates, this closed-shoe paradigm fails catastrophically when applied within the socioeconomic and environmental conditions of India[1].

Comprehensive scoping reviews analyzing diabetic self-care and footwear practices in low- and middle-income countries demonstrate that between 70% and 87% of diabetic individuals in India routinely wear open sandals, slippers, or traditional chappals, with less than 10% to 16% regularly utilizing closed footwear[1]. Tropical ambient temperatures and monsoon humidity make enclosed footwear intolerable, inducing severe hyperhidrosis and skin maceration[19]. When insensate diabetic feet are trapped within humid, enclosed environments, cutaneous fungal infections such as tinea pedis proliferate rapidly[19]. Clinical studies analyzing lower-extremity ulcer depth indicate that closed-shoe wearers experience fungal infections at elevated frequencies (30.5%) compared to open-footwear users (25.3%), creating epidermal fissures that serve as entry portals for invasive bacterial pathogens and subsequent limb-threatening osteomyelitis[19].

Furthermore, deep-rooted cultural customs dictate that footwear be removed prior to entering domestic residences and places of worship[1]. In these indoor spaces, patients navigate completely barefoot or in thin, unsupportive bathroom slippers[1]. Epidermal trauma occurring during unprotected indoor ambulation is one of the leading predictors of advanced ulcer depth and subsequent surgical amputation in Indian diabetic cohorts[17]. Prescribing rigid, expensive enclosed leather shoes under these conditions frequently results in non-compliance, leaving patients unprotected during their most active daily hours[1].

Indian Institutional and Academic Solutions

Recognizing the clinical hazard of importing Western footwear mandates into the Indian subcontinent, premier Indian scientific institutions—including the CSIR-Central Leather Research Institute (CSIR-CLRI), the Indian Institute of Science (IISc), and specialized diabetes research hospitals—have developed therapeutic and instrumented footwear designed specifically for open-shoe form factors[6].

At CSIR-CLRI in Chennai, research teams collaborating with the Prof. M. Viswanathan Diabetes Research Centre (MVDRC) focused extensively on developing biomechanically optimized open footwear[6]. This work culminated in the patenting and clinical deployment of DIASTEP (Indian Patent No. 302551), an indigenous therapeutic footwear system featuring dual-density microcellular rubber (MCR) insoles, polyurethane outsoles, and multi-strap retention systems[6]. In parallel clinical trials led by Dr. Vijay Viswanathan, researchers evaluated indigenous offloading sandals constructed with rigid rocker soles and compliant footbeds against imported, high-cost Western offloading boots[16]. The clinical data demonstrated that indigenous open offloading sandals were statistically equivalent to expensive foreign devices in promoting diabetic foot ulcer healing and preventing recurrence, while achieving significantly higher patient compliance[16]. To validate these interventions, CSIR-CLRI integrates pedobarographic sensor arrays, such as the Podoprint pressure mat, to map plantar pressure redistribution across prototype sandal soles and therapeutic insoles directly[22].

At the Indian Institute of Science (IISc) Bengaluru, a multidisciplinary research team led by Prof. G.K. Ananthasuresh from the Department of Mechanical Engineering and Dr. Pavan Belehalli from the Karnataka Institute of Endocrinology and Research (KIER) engineered dynamic self-offloading therapeutic footwear[9]. Published in *Wearable Technologies*, their work addresses the fundamental limitations of static insoles and expensive, fragile electronic sensor-actuator networks by utilizing the principles of compliant mechanisms[9]. The team designed and additively manufactured 3D-printed TPU footbeds embedded with an array of bistable snap-through arches[9].

The mechanical architecture of the IISc system functions simultaneously as a sensor and an actuator[9]. Each arch is computationally dimensioned according to the patient's body mass, gait velocity, and anatomical pressure profile[9]. When dynamic ground contact forces exceed a critical threshold—calibrated around a clinical limit of $200\ \mathrm{kPa}$ to prevent tissue ischemia—the compliant arch snaps into a negative-stiffness regime, immediately offloading the vulnerable site and redistributing mechanical stresses to surrounding healthy tissue[9].

Clinical evaluations conducted on neuropathic subjects demonstrated a 57% peak pressure reduction in the forefoot[9].

Crucially, because this mechanical intelligence is embedded entirely within the elastomeric lattice of the footbed, it operates without external power supplies, fragile wire harnesses, or environmental encasements, making it fully compatible with open sandals and slippers[9]. This platform was integrated into the Digital Podiatry Clinic (D-PoC) initiative under the Bengaluru Science and Technology (BeST) cluster and translated into clinical adoption through health-technology enterprises including StrideAide, FootSecure, and Yostra Labs[15].

Academic engineering institutions across India have similarly explored electronic sensing within open footwear[26]. Investigators such as Patle et al. constructed IoT-enabled smart sandals integrating Arduino microcontrollers, tactile force sensors, Bluetooth Low Energy modules, and GSM/GPS transceivers within open footwear bases[27]. Talpur and co-workers developed sensor-instrumented open sandal platforms combining local tactile switches and telemetry modules[26]. While several academic projects focused on remote safety tracking rather than chronic ulceration, their underlying hardware design solved the electromechanical packaging requirements of routing data links and power lines through open sandal chassis[26].

Translational Medical Devices and the Commercial Smart Sandal Ecosystem

The assertion that no product has addressed the sensorization of open footwear for diabetic care is directly contradicted by specialized medical device startups and established clinical suppliers[5].

The most prominent dedicated smart sandal platform engineered specifically for diabetic neuropathy is SensorixFoot, an active medical device developed to overcome the clinical failure modes of enclosed diabetic footwear[5]. Conceptualized by clinical nursing and engineering specialists, SensorixFoot incorporates continuous thermal monitoring and localized mechanical therapy within an open, breathable sandal architecture[5].

The primary sensor array comprises calibrated thermal sensors embedded across key plantar vascular control zones and beneath the dorsal instep strap[5]. Because continuous focal hyperthermia ($\geq 2.2^{\circ}\mathrm{C}$ or $4^{\circ}\mathrm{F}$) serves as an objective physiological precursor to subcutaneous inflammation and tissue breakdown, the system provides early warnings of impending ulceration prior to macroscopic skin necrosis[5]. To eliminate the discomfort and infection risks of closed shoes, the sandal incorporates an open-toe upper with specialized exposure windows over sensing nodes, ensuring accurate dermal contact while maintaining airflow[5].

Beyond passive monitoring, the sandal integrates therapeutic vibration modules that deliver targeted mechanical stimulation to cutaneous mechanoreceptors, promoting peripheral blood flow, stimulating tissue perfusion in ischemic microvascular beds, and alleviating sensory deficits associated with diabetic neuropathy[5]. To ensure durability in real-world environments without delicate charging ports, the system features a hermetically sealed, IP68-rated wireless charging receiver embedded in the lower sole, recharging via a contactless inductive floor

mat[5]. Dynamic sensor readings are processed via an onboard Bluetooth Low Energy module and transmitted to a mobile application using the "Smart Predictive Alert" (S.P.A.) protocol, which employs artificial intelligence algorithms to categorize tissue status into a tri-color risk tier (green, yellow, red) and generate automated diagnostic summaries for attending podiatrists[5].

Within the domestic Indian market, a commercial supply chain exists to provide specialized diabetic footwear incorporating microcellular polymer (MCP), microcellular rubber (MCR), and silicone gel footbeds within open sandal designs[29]. Commercial manufacturers—such as Dia One, Biolimb Hi-Tech Rehab Center, and Hochste Healthcare—supply multi-strap orthopedic sandals constructed with 8 mm to 10 mm low-shore MCP insoles, contoured arch profiles, and dual-density shock-absorbing outsoles[29]. These devices are specifically prescribed to redistribute plantar pressures across the footbed while preserving the open, ventilated architecture required for compliance in tropical Indian settings[29].

Biomechanical and Mechatronic Engineering Constraints

The historical concentration of commercial smart insoles in closed shoes is rooted in severe electromechanical engineering hurdles, rather than a failure to recognize the global reliance on open footwear[1]. Resolving these issues in a sandal requires solving distinct structural and telemetry challenges[5].

In an enclosed shoe, the heel counter, vamp, and rigid toe box mechanically stabilize the foot, constraining relative planar translation between the plantar skin and the insole surface[1]. In an open sandal or chappal, the lack of dorsal containment induces substantial longitudinal and transverse shear forces during late midstance and push-off[36]. These shear stresses not only elevate the risk of blister formation and ulceration in insensate feet, but also subject surface-mounted sensor substrates to destructive mechanical delamination[7]. Sandal systems resolve this by incorporating contoured footbeds, deep calcaneal heel cups, anatomical toe crests, and multi-point adjustable strapping configurations that stabilize the skeletal framework without enclosing the foot[35].

Environmental sealing presents another critical challenge. While closed shoes provide an internal barrier that buffers the sensor package against dust, moisture, and mechanical impact, smart sandals operate in direct contact with rain, abrasive mud, and street debris[5]. Routing delicate flex-circuit cables through an open sandal sole frequently induces rapid wire fatigue at dynamic flex points[2]. Engineers have overcome these vulnerabilities by transitioning toward continuous polymer optical fibers, carbon nanotube-impregnated nonwoven fabrics, fully encapsulated rigid-flex printed circuit boards potted in waterproof polyurethane resins, and inductive charging arrays that eliminate vulnerable external connectors[2].

Furthermore, thermal baseline drift introduces algorithmic complexities in open devices[5]. In an enclosed shoe, the internal microclimate stabilizes within a narrow thermodynamic envelope, providing a consistent baseline for differential thermometry[39]. In contrast, open sandals are exposed to ambient airflow, wind fluctuations, and solar radiation[35]. Device architectures like

SensorixFoot counteract this thermal variability by utilizing comparative, multi-site algorithms that correlate contralateral plantar data against non-weight-bearing dorsal reference sensors on the sandal upper, isolating genuine pathologic hyperthermia from environmental cooling[5].

Engineering Dimension	Standard Enclosed Smart Insole	Instrumented Sandal System	Technical Solution in Sandal Literature
Foot Fixation & Shear Dynamics	Heel counter, toe box, and rigid upper prevent foot slippage[1].	Unconstrained borders allow significant longitudinal and lateral shear[36].	Multi-strap adjustable retention, deep heel cups, and contoured arch footbeds[35].
Hardware Packaging & Enclosure	Electronics sheltered within deep midsole cavities and heel wedges[40].	Thin sole profile exposes packaging to lateral impact and bending.	Rigid-flex PCBs embedded in molded EVA/PU soles or strap anchor posts[5].
Environmental Ingress	Enclosed upper shields insole from direct rain, mud, and debris.	Full exposure to environmental water, high humidity, and abrasive dirt.	Hermetic IP68 polymer potting and contactless Qi/inductive charging coils[5].
Wire Fatigue & Flex Stress	Wiring routes safely along internal shoe wall linings[2].	Repetitive dynamic flexing across unsupported borders shears connections.	Continuous flexible optical fibers, conductive CNT textiles, and molded conduits[2].
Thermal Sensing Baseline	Enclosed cavity reaches an insulated, quasi-stable thermal microclimate[39].	Ambient airflow and solar exposure generate fluctuating surface baselines.	Differential algorithms cross-referencing contralateral zones and dorsal strap sensors[5].

Nuanced Assessment of the Original Assertion

An objective evaluation of the statement highlights a clear dichotomy between public health observation and technological reality:

The assertion correctly captures the podiatric public health critique regarding the mismatch between Western medical guidelines and the lifestyle realities of tropical, low-resource nations[1]. The vast majority of commercially marketed consumer smart insoles (such as FeetMe, Digitsole, and Tekscan Pedar) were indeed developed for closed athletic or orthopedic shoes, reflecting the climatic and market conditions of Western nations[2]. Prescribing closed therapeutic shoes to India's rural and urban diabetic patients frequently induces non-compliance, hyperhidrosis, and fungal infections, failing to protect patients who perform the majority of their daily steps barefoot or in open footwear[1].

However, the categorical claim that *"no sensor, no product, no paper has ever addressed this"* across any database globally is completely unfounded. A substantial, highly active body of scientific research and medical technology development directly addresses this challenge[2]. Peer-reviewed studies in journals including *Sensors*, *Wearable Technologies*, and the *International Journal of 3D Printing Technologies and Digital Industry* have documented functional, sensorized smart sandals utilizing polymer optical fibers, carbon nanotube textile piezoresistors, and FSR matrices[2]. Premier Indian research institutions—notably CSIR-CLRI and the Indian Institute of Science—have developed, patented, and commercialized therapeutic rocker sandals and dynamically self-offloading snap-through footbeds designed specifically for non-closed footwear[6]. Commercial smart devices like SensorixFoot integrate active multi-zone thermal sensors, therapeutic mechanoreceptor stimulators, and IP68 wireless charging into intelligent sandals to protect neuropathic diabetic patients[5].

Far from being an ignored problem, adapting advanced sensing, mechanical metamaterials, and wireless telemetry to open-footwear form factors represents an established, highly sophisticated discipline within contemporary biomedical engineering[5].

Works cited

1. A Scoping Review of Footwear Worn by People With Diabetes in Low, https://pmc.ncbi.nlm.nih.gov/articles/PMC10141434/
2. FORCE ESTIMATION IN SMART INSOLES USING RANDOMLY, https://dergipark.org.tr/tr/download/article-file/5243108
3. Polymer Optical Fiber Plantar Pressure Sensors: Design and ... - MDPI, https://www.mdpi.com/1424-8220/22/10/3883
4. Sandal with nanocomposite sensors on the hindfoot and forefoot, https://www.researchgate.net/figure/Sandal-with-nanocomposite-sensors-on-the-hindfoot-and-forefoot-regions-of-the-insole_fig1_368967795
5. Smart care for yur feets - Sensorixfoot, https://www.sensorixfoot.cl/en
6. Footwear for Diabetic Patients - CSIR, https://www.csir.res.in/en/csir-success-stories/footwear-diabetic-patients
7. Recent Advances in 3-D Printed, Wearable Pressure Sensors for, https://www.researchgate.net/publication/384285235_Recent_Advances_in_3-D_Printed_Wearable_Pressure_Sensors_for_Plantar_Pressure_Monitoring_A_Review

8. Design and Development of a High Resolution Open-Source Plantar, https://www.hilarispublisher.com/open-access/design-and-development-of-a-high-resolution-opensource-plantar-pressure-sensing-system-76148.html
9. Self-offloading therapeutic footwear using compliant snap-through, https://pmc.ncbi.nlm.nih.gov/articles/PMC10936371/
10. A Technological Review of the Instrumented Footwear for ... - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC4719096/
11. Carmen Vázquez's research works | University Carlos III de Madrid, https://www.researchgate.net/scientific-contributions/Carmen-Vazquez-2149919878
12. Estimating ground reaction force with novel carbon nanotube-based, https://pmc.ncbi.nlm.nih.gov/articles/PMC10062471/
13. force estimation in smart insoles using randomly placed optical fiber, https://www.researchgate.net/publication/399131664_FORCE_ESTIMATION_IN_SMART_INSOLES_USING_RANDOMLY_PLACED_OPTICAL_FIBER_SENSORS_AND_MACHINE_LEARNING
14. A Technological Review of the Instrumented Footwear for ... - Frontiers, https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2016.00001/full
15. Digital podiatry clinic - Footprints of progress - IndiaBioscience, https://indiabioscience.org/columns/indian-scenario/digital-podiatry-clinic-footprints-of-progress
16. School of Rock(er): How shoe modifications can make a big difference, https://diabeticfootonline.com/2011/08/24/school-of-rocker-how-shoe-modifications-can-make-a-big-difference/
17. IWGDF Guidelines on the prevention and management of diabetic, https://iwgdfguidelines.org/wp-content/uploads/2019/05/IWGDF-Guidelines-2019.pdf
18. IWGDF Guidelines on the prevention and management of diabetes, https://iwgdfguidelines.org/wp-content/uploads/2023/07/IWGDF-Guidelines-2023.pdf
19. Time spent barefoot predicts diabetic foot ulcer depth, https://diabetesonthenet.com/diabetic-foot-journal/time-spent-barefoot-predicts-diabetic-foot-ulcer-depth/
20. About Us - strideAide, https://www.strideaide.com/aboutus.html
21. CSIR Annual Report, https://www.csir.res.in/sites/default/files/2023-06/Annual%20Report%202009-2010.pdf
22. cum-Convention Centre - 'Bharat Mandapam', July 26, 2023, https://leatherindia.org/wp-content/uploads/2023/09/LEATHER-NEWS-INDIA-AUGUST-2023.pdf
23. 'Snapping' footwear to help prevent diabetic foot complications, https://www.iisc.ac.in/events/snapping-footwear-to-help-prevent-diabetic-foot-complications/
24. Bengaluru: Researchers develop snapping footwear to help prevent,

https://indianexpress.com/article/cities/bangalore/bengaluru-researchers-develop-snapping-footwear-diabetic-foot-7967765/
25. IISc develops footwear to help prevent diabetics foot complications, https://www.hindustantimes.com/india-news/iisc-develops-footwear-to-help-prevent-diabetics-foot-complications-101655146927450.html
26. (PDF) Smart Sandal with IoT for Women's Safety and Empowerment, https://www.researchgate.net/publication/403592561_Smart_Sandal_with_IoT_for_Women's_Safety_and_Empowerment
27. Design and Development of IoT Enabled Smart Sandal, https://ijarsct.co.in/A3351.pdf
28. Design and Development of IoT Enabled Smart Sandal, https://ijarsct.co.in/Paper3351.pdf
29. Leather Silicone Insole Diabetic Sandal at ₹ 5000/pair - IndiaMART, https://www.indiamart.com/proddetail/leather-silicone-insole-diabetic-sandal-15474153033.html
30. About Us - Sensorixfoot, https://www.sensorixfoot.cl/en/about
31. Venezuelan creates sandals to prevent amputations due to diabetes, https://www.youtube.com/watch?v=nXbqa3hGFMc
32. SENSORIX FOOT: The Smart Therapeutic Sandal - Create the Future, https://contest.techbriefs.com/2025/entries/medical/13812-0704-122051-sensorix-foot-the-smart-therapeutic-sandal
33. Dia One Orthopedic Sandal PU Sole MCP Insole Diabetic Footwear, https://www.1mg.com/otc/dia-one-orthopedic-sandal-pu-sole-mcp-insole-diabetic-footwear-for-men-and-women-dia-53-size-6-otc402381
34. Dia One Orthopedic Sandal Rubber Sole Mcp Insole Pain Relief, https://www.desertcart.in/products/124806472-orthopedic-sandal-rubber-sole-mcp-insole-pain-relief-diabetic-footwear-ladies-chappal-for-women-dia_96_p
35. Expert Guide to Buying Durable Women's Universal Strap Sandals, https://world.taobao.com/lang/en-us/shopping-guide/2050958423939350528.htm
36. Sandals: How to Choose Supportive Styles That Won't Hurt Your Feet, https://denielfootandanklecenter.com/sandals-how-to-choose-supportive-styles-that-wont-hurt-your-feet/
37. How to Find Sandals That Truly Fit: 7 Checks for All-Day Comfort, https://projectcloudshoes.com/blogs/faq-blog-news/7-sandal-fit-checks-comfort-guide
38. Women Removable Sandal at Jem, https://www.jem.sg/whats-new/women-removable-sandal3/
39. Relationship between Plantar Pressure and Sensory Disturbance in, https://pmc.ncbi.nlm.nih.gov/articles/PMC7730212/
40. Footwear | IEEE Technology Navigator, https://technav.ieee.org/topic/footwear/